

Epithelial Tissue

Department of Anatomy

• Histology(2)

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Lecturer points:

- Most organs can be divided into?
(Parenchyma and Stroma).
- Structure of Epithelia.
- Basement Membranes.
 - Example: kidney glomeruli the basement membrane
- Intercellular Adhesion & Other Junctions.
- The principal functions of epithelial tissues.
- Epithelia can be divided into two main groups.
 - **Covering (or lining) epithelia.**
 - **Secretory (glandular) epithelia.**
- Covering (or lining) epithelia.
- Classification of covering or Lining Epithelia.
- Types of epithelia.
- Common types of covering epithelia

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Parenchyma:

- which is composed of the cells responsible for the organ's specialized functions.

Stroma:

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- Except in the brain and spinal cord, the stroma is always connective tissue.

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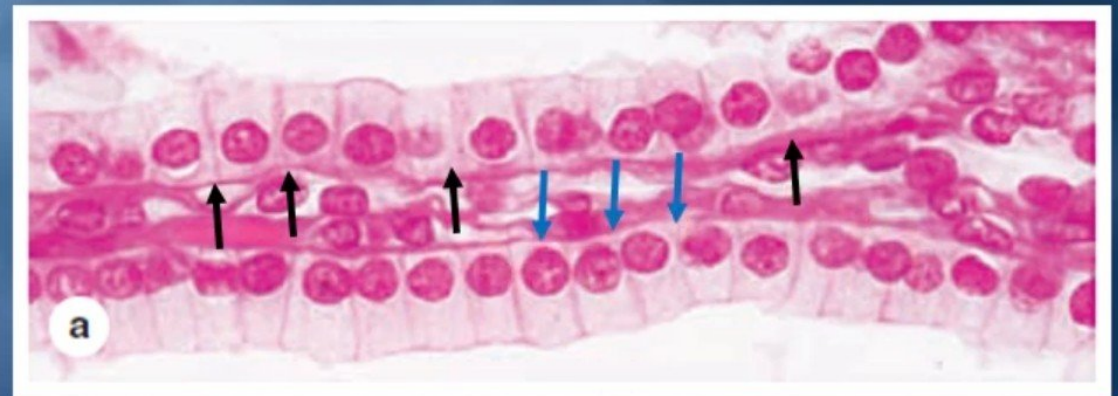
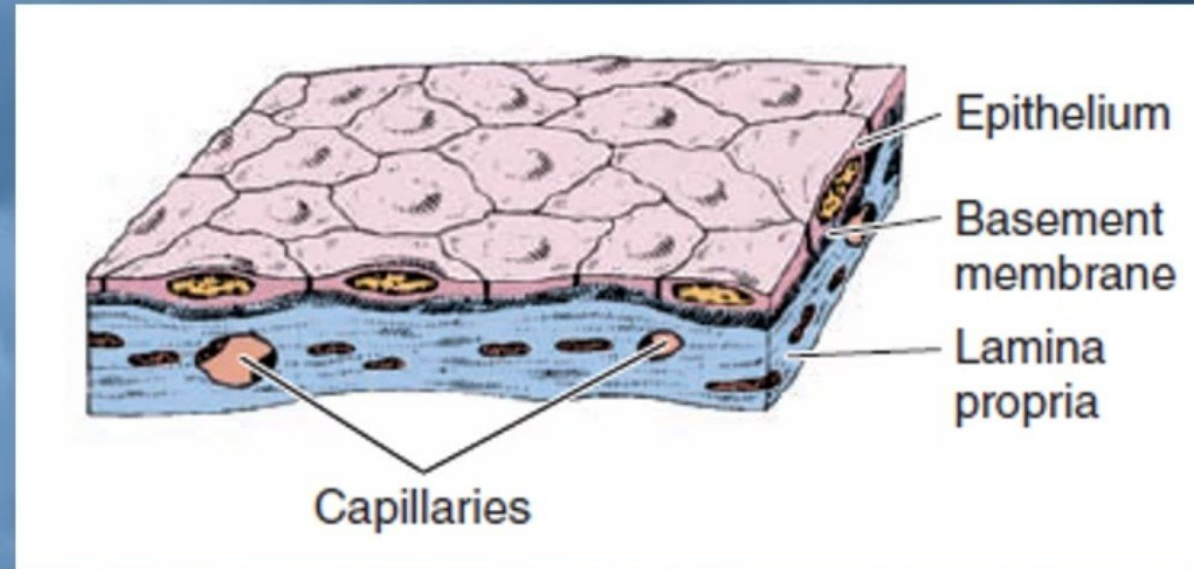
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Structure of Epithelium

- They are specialized layers of tissue **arising from all three embryonic germ layers**, (the **ectoderm**, **mesoderm**, and **endoderm**) that **line the internal** and **cover the external surfaces of the body** except in certain areas such as tooth surfaces and articular cartilages.
- An epithelium consists of a sheet of **cells lying close together** with **little extracellular space**.
- **The basement membrane:**
 - composed of a **basal lamina** and a **lamina reticularis**, **separates the epithelium** from underlying **connective tissue** and **blood vessels**.
- Epithelia are **avascular** and receive **nourishment** by **diffusion** of molecules through **the basal lamina** to which they are attached.

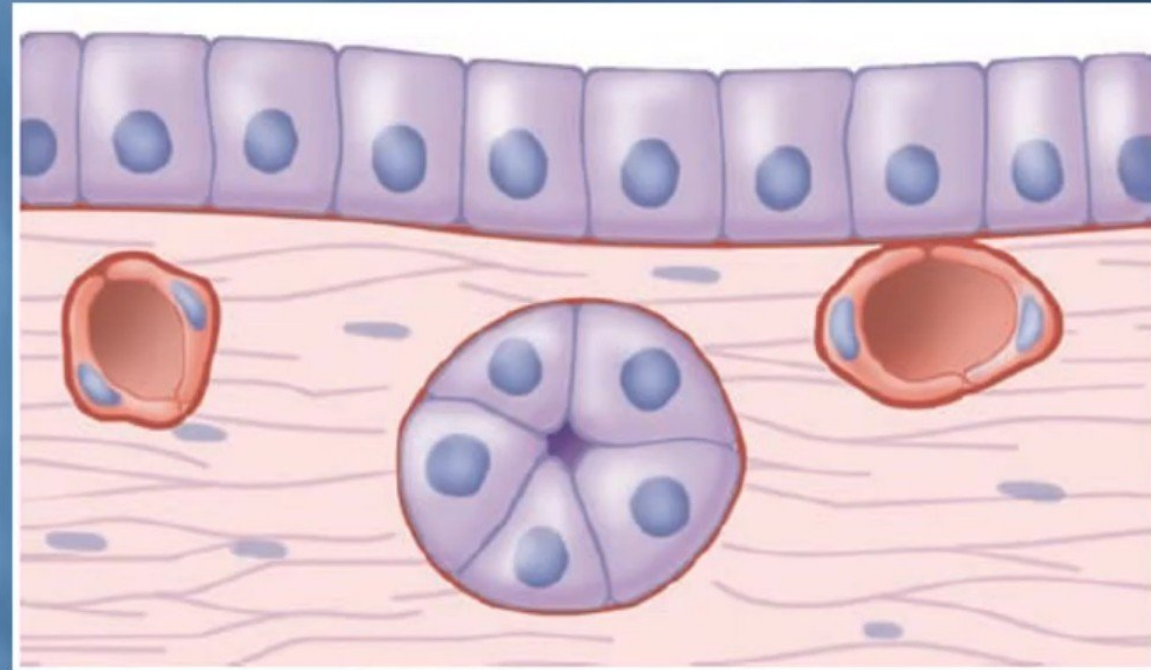
Basement Membranes

- All epithelial cells in contact with subjacent **connective tissue** have at their **basal surfaces** a specialized, **felt-like sheet of extracellular material** referred to as the basement membrane.
- **Glycoproteins** and other components in this structure can be **stained** and **make it visible beneath epithelia** with the light microscope.



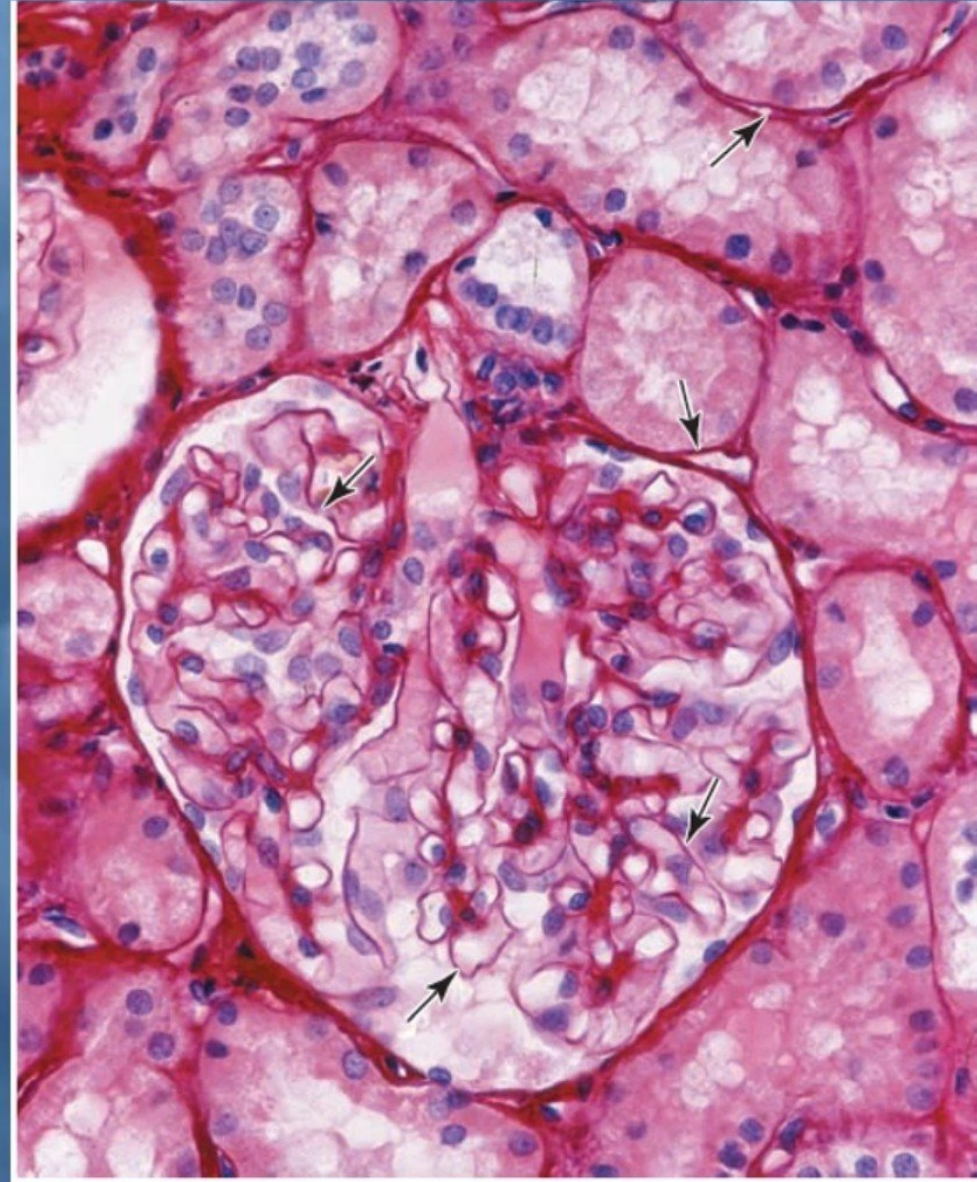
Basement Membrane

- **Cuboidal** or **pyramidal cells** of epithelia generally have **spherical nuclei**, while **nuclei of squamous epithelial cells** are **flattened**.
- An **extracellular basement membrane (red)** always lies at the interface of epithelial cells and **connective tissue**.
- **Nutrients** for epithelial cells must **diffuse across** the **basement membrane**.
- **Nerve fibers** normally penetrate this structure, but **small blood capillaries** (being epithelial themselves) normally never **enter epithelia**.



kidney glomeruli the basement membrane

- This section of **kidney** shows the **well-stained basement membranes (arrows)** of epithelia forming structures within the large, **round renal glomerulus** and its **surrounding tubules**.
- In kidney glomeruli the basement membrane, besides having a **supporting** function, has a highly developed **role as a filter** that is key to renal function.
- X100. Picro sirius hematoxylin (PSH).

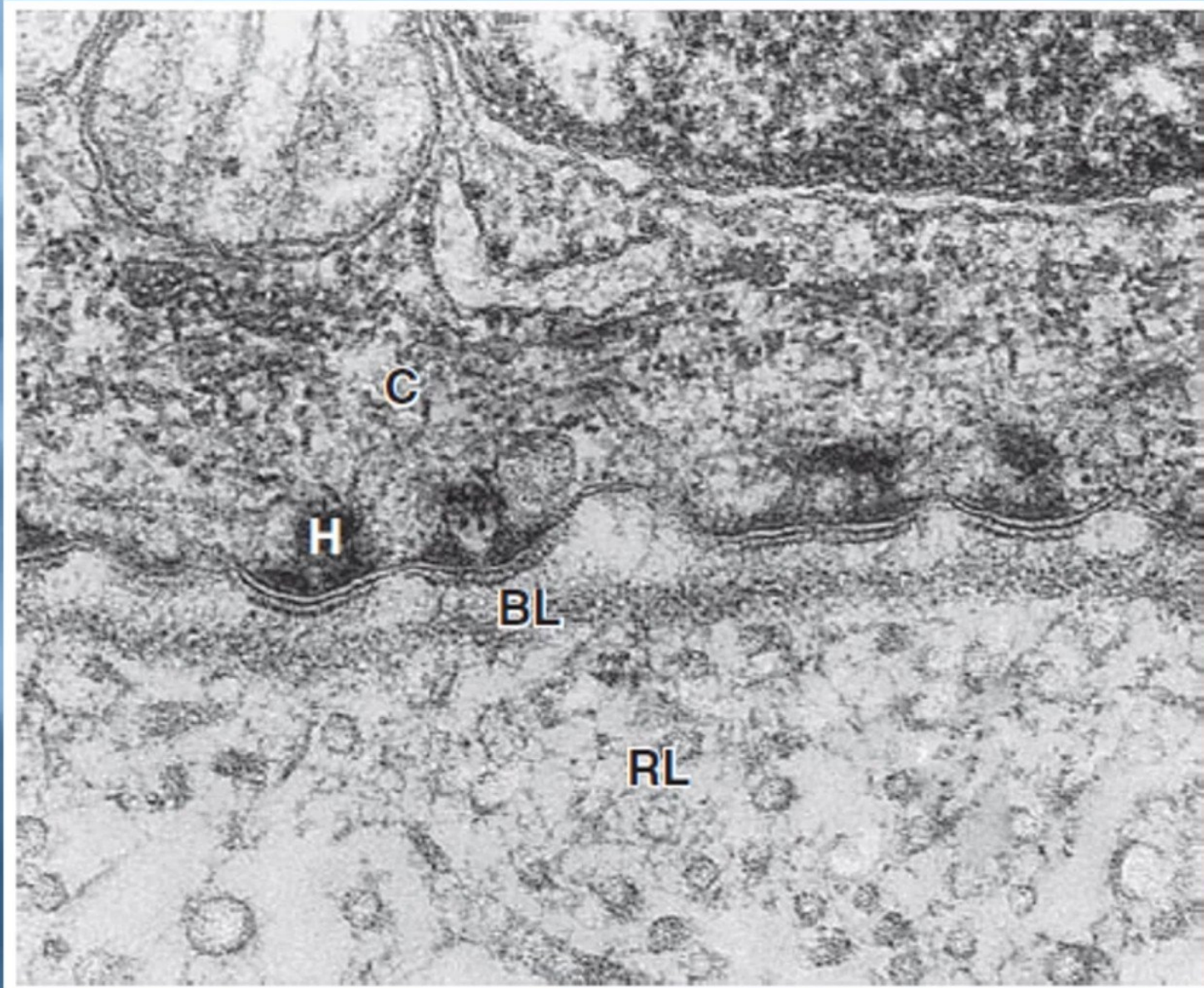


With the **transmission electron microscope** (TEM) the **basement membrane may be resolved into two structures:**

- Nearest the epithelial basal poles is an electron-dense layer, **20-100 nm thick**, consisting of a network of fine fibrils that comprise **the basal lamina (BL)**.
- Beneath this layer is often a more diffuse and **fibrous reticular lamina (RL)**.

Note:

- **Hemidesmosomes (H)** bind the basal surface of the **epithelial cell (C)** to the basal lamina.



Intercellular Adhesion & Other Junctions

Epithelial cells adhere strongly to neighboring cells and basal laminae.

Junction	Tight Junction (Zonula Occludens)	Adherent Junction (Zonula Adherens)	Desmosome (Macula Adherens)	Hemidesmosome	Gap Junction (Nexus)
Major transmembrane link proteins	Occludins, claudins, ZO proteins	E-cadherin, catenin complexes	Cadherin family proteins (desmogleins, desmocollin)	Integrins	Connexin
Cytoskeletal components	Actin filaments	Actin filaments	Intermediate filaments (keratins)	Intermediate filaments	None
Major functions	Seals adjacent cells to one another, controlling passage of molecules between them; separates apical and basolateral membrane domains	Provides points linking the cytoskeletons of adjacent cells; strengthens and stabilizes nearby tight junctions	Provides points of strong intermediate filament coupling between adjacent cells, strengthening the tissue	Anchors cytoskeleton to the basal lamina	Allows direct transfer of small molecules and ions from one cell to another
Medical significance	Defects in occludins may compromise the fetal blood-brain barrier, leading to severe neurologic disorders	Loss of E-cadherin in epithelial cell tumors (carcinomas) promotes tumor invasion and the shift to malignancy	Autoimmunity against desmoglein I leads to dyshesive skin disorders characterized by reduced cohesion of epidermal cells	Mutations in the integrin- β 4 gene are linked to some types of epidermolysis bullosa, a skin blistering disorder	Mutations in various connexin genes have been linked to certain types of deafness and peripheral neuropathy

The principal functions of epithelial tissues include the following:

- 1) **Covering, lining**, and protecting surfaces (eg, epidermis).
- 2) **Absorption** (eg, the intestinal lining).
- 3) **Secretion** (eg, parenchymal cells of glands).
- 4) **Specific cells** of certain epithelia: may be
 - **Contractile (myoepithelial cells)**.
 - **Specialized sensory cells**, such as those of **taste buds** or the **olfactory epithelium**.
- 5) **Transcellular transport** of molecules from one epithelial surface to another occurs by various processes, including the following:
 - a) **Diffusion** of oxygen and carbon dioxide across the epithelial cells of lung alveoli and capillaries.
 - b) **Carrier protein mediated** transport of amino acids and glucose across intestinal epithelia
 - c) **Vesicle mediated transport** of immunoglobulin A (IgA) and other molecules.

Thanks